

Proposal for Demyelinating Disease Classification

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1 Background Information

1.1 Key Terms

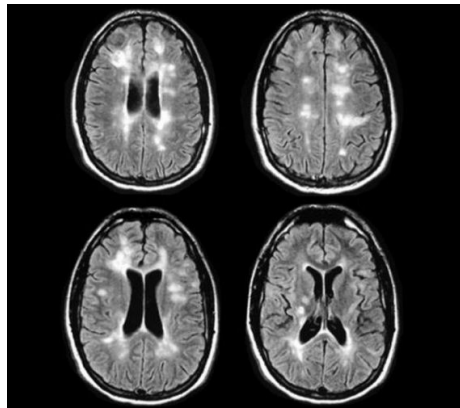


Figure 1: Example of MS lesions

Multiple Sclerosis: A demyelinating disease which presents with brain lesions which can be seen on an MRI. The most common form of demyelinating illness. Abbreviated as MS.

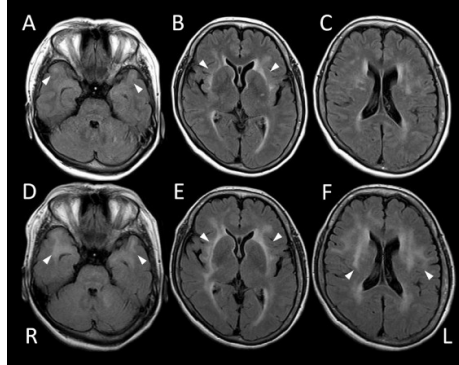


Figure 2: Example of NMOSD lesions

Neuromyelitis Optica Spectrum Disorder: Another demyelinating disease which presents with brain lesions which can be seen on an MRI. Much less common than MS. Abbreviated as NMOSD.

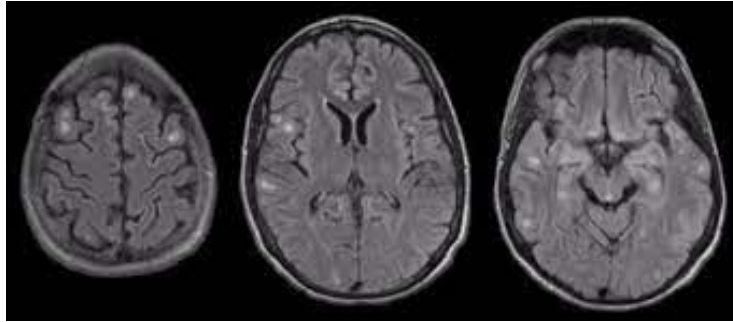


Figure 3: Example of ADEM lesions

Acute Disseminated Encephalomyelitis: Another demyelinating disease which presents with brain lesions which can be seen on an MRI. As common as NMOSD. Abbreviated as ADEM.

1.2 Prior Research

Within the past 5 years, demyelinating disease classification using ML has become a prominent topic in the field of computational biology. In 2020, Wang et. al developed the first break through in demyelinating disease classification. In this influential paper, a method of differentiating MS from NMOSD based on 3 dimensional Brain MRI images using pure convolutional neural networks was detailed. This relatively simple model achieved an accuracy between 0.725 and 0.75 (Wang et al., 2020). A second breakthrough was reached when Zhou et. al passed MRI images into a U-Net segmentation algorithm to separate the

healthy matter from the lesions before putting the images through a CNN. In this study, they differentiated NMOSD from ADEM with an accuracy of 95.55% (Zhou et al., 2024).

2 Proposal

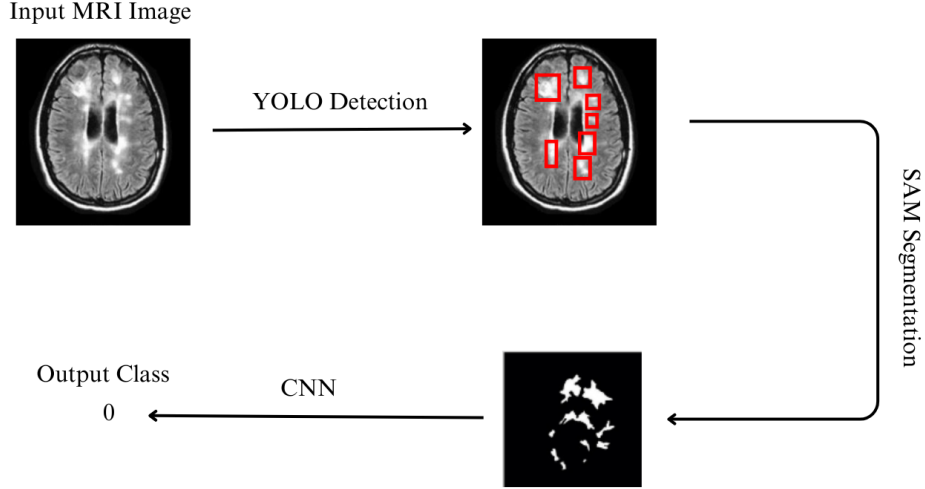


Figure 4: Proposed Pipeline for classification model to differentiate NMOSD from MS

We can introduce novelty to produce a result similar to Zhou et. al in 2 distinct ways. First, rather than differentiating NMOSD with ADEM, we can aim to differentiate NMOSD with MS or ADEM with MS. I suggest that we focus on NMOSD differentiation due to data availability. Second, in our differentiation pipeline, we can utilize Meta’s new Segment Anything Model rather than the dated U-Net segmentation model (Kirillov et al., 2023). Our novel pipeline will hopefully be able to differentiate NMOSD from MS with an accuracy similar to that achieved in Zhou et. al’s paper on differentiating NMOSD from ADEM (Zhou et al., 2024). In order to achieve significant results, we should aim for an accuracy above 75% as to surpass the accuracy achieved by Wang’s model (Wang et al., 2020).

Bibliography

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Wang, Z., Yu, Z., Wang, Y., Zhang, H., Luo, Y., Shi, L., Wang, Y. and Guo, C. (2020). 3D Compressed Convolutional Neural Network Differentiates Neuromyelitis Optic Spectrum Disorders From Multiple Sclerosis Using Automated White Matter Hyperintensities Segmentations. *Frontiers in physiology*, 11. doi:<https://doi.org/10.3389/fphys.2020.612928>.

Zhou, D., Xu, L., Wang, T., Wei, S., Gao, F., Lai, X. and Cao, J. (2024). M-DDC: MRI based demyelinating diseases classification with U-Net segmentation and convolutional network. *Neural networks*, 169, pp.108–119.
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